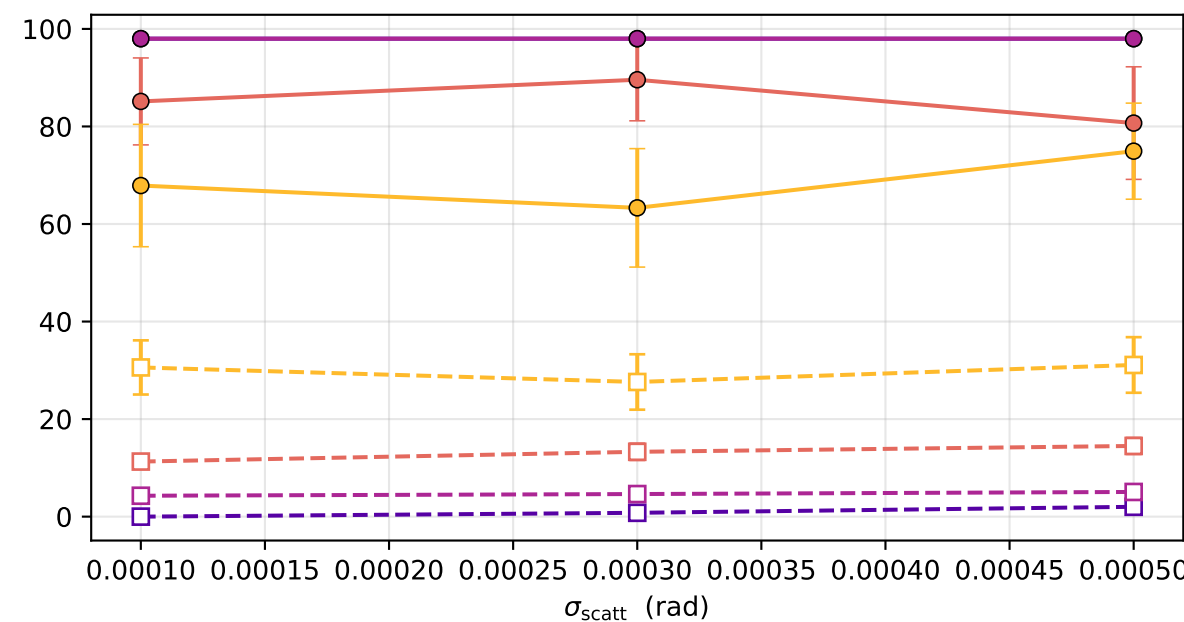
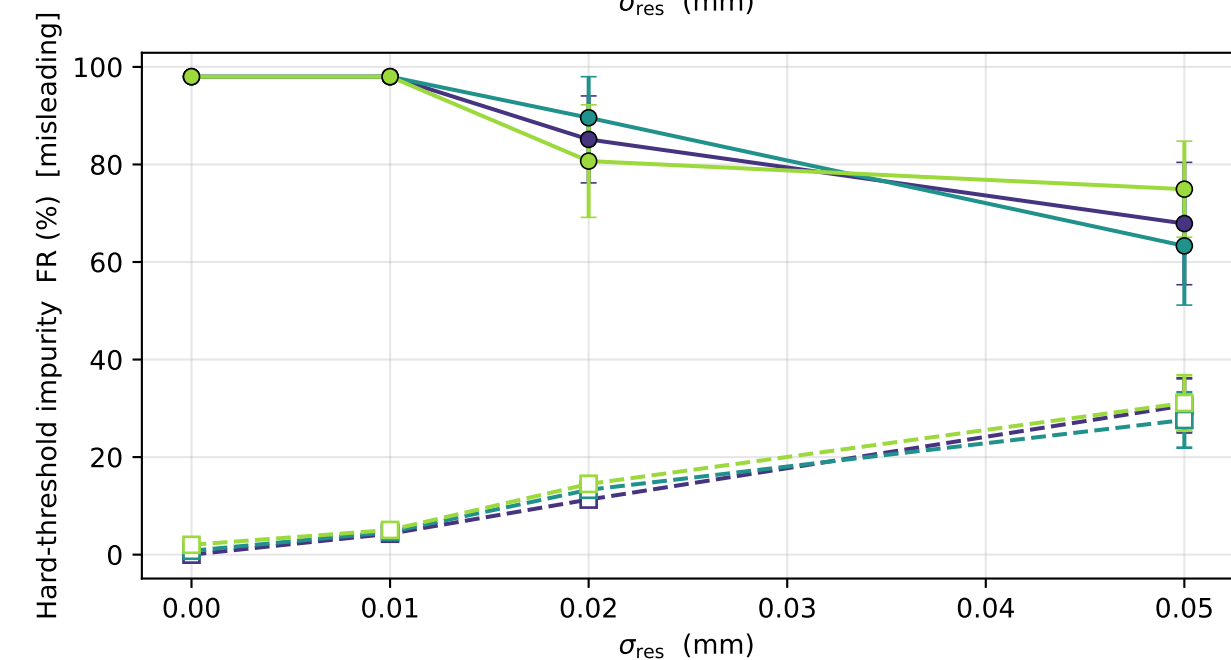
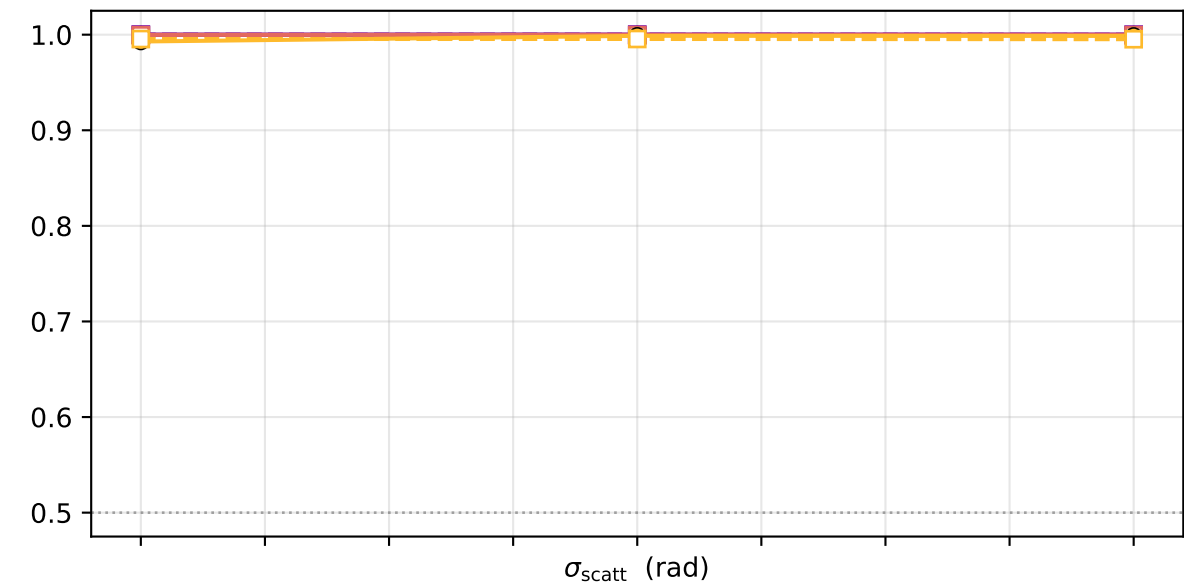
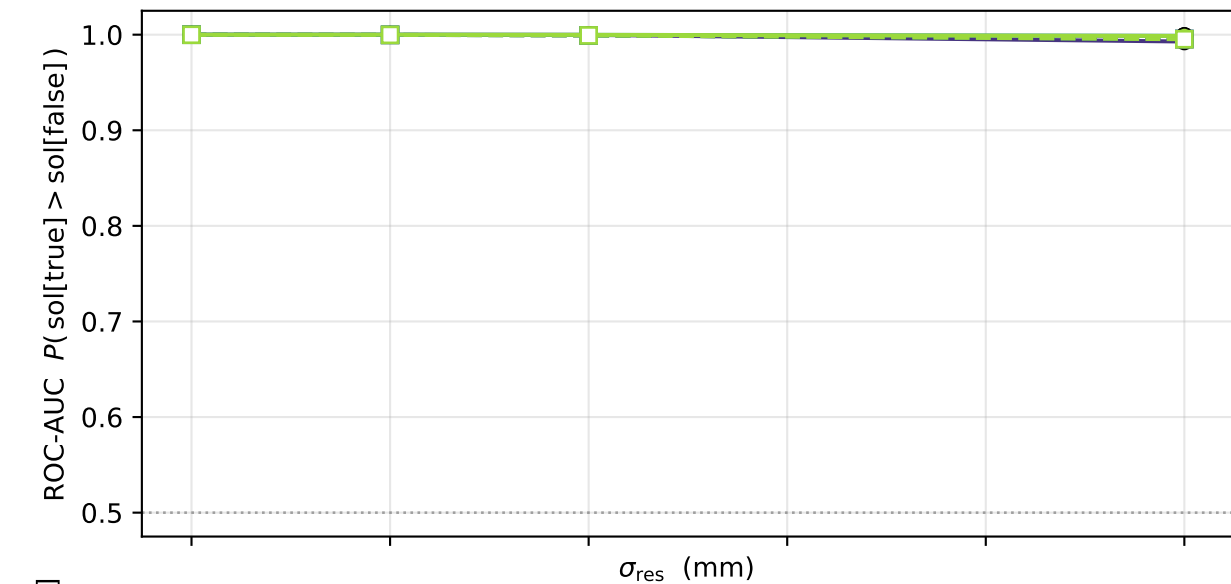
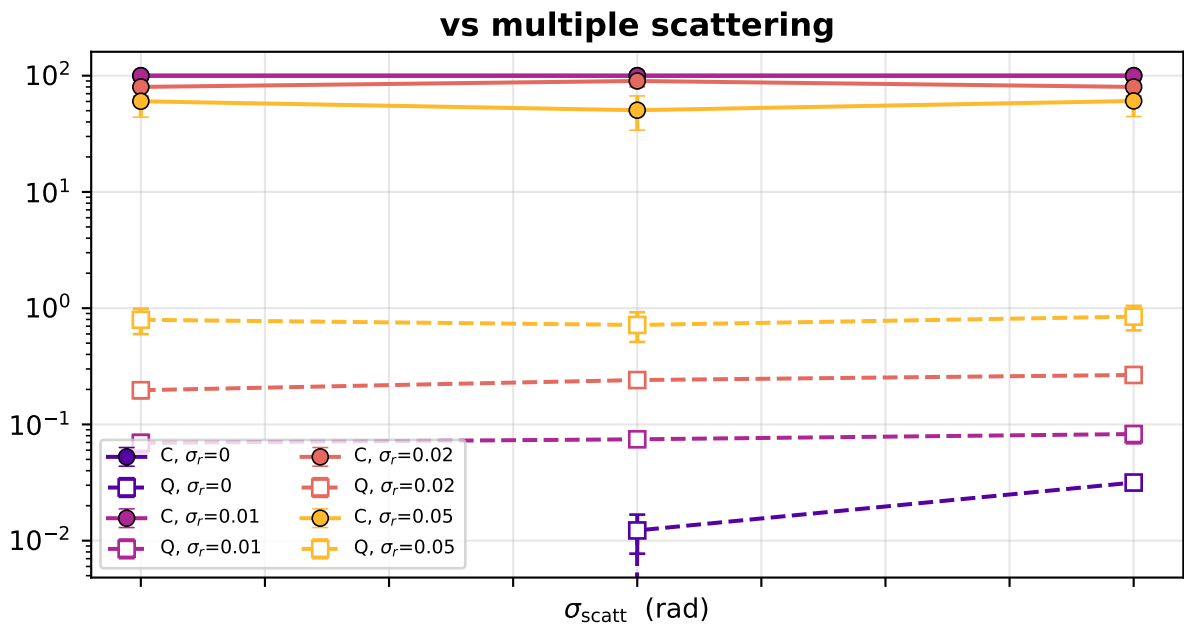
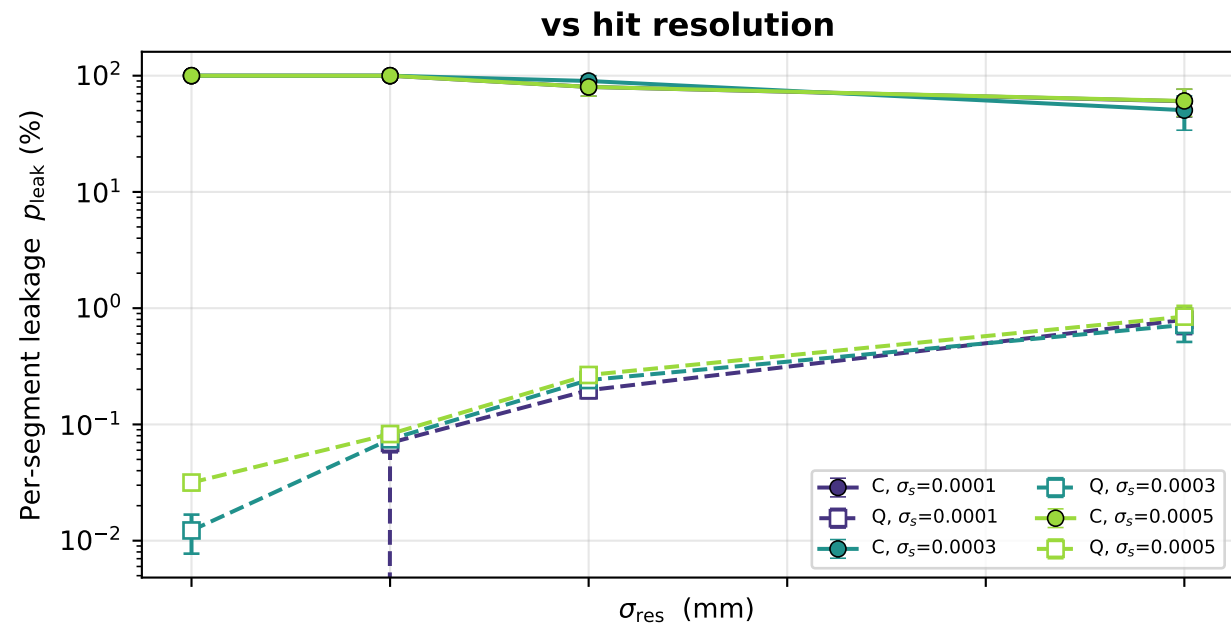


Threshold-robust segment metrics — $T = 50$ tracks (solid = classical, dashed = 1-bit-HHL)



Controls: $\gamma=3$, $\delta=1$, $\phi_{\text{max}}=\theta_{\text{max}}=0.2$ rad, $\theta_{\text{min}}=1.5\text{e-}5$, $\tau=0.35$, $\varepsilon=\text{calc.}(s=3)$, conv OFF, hit-ineff 0, 5 planes $\Delta z=33$ mm, statevector 1BQF
 Fixed here: per-segment leakage & impurity at $\tau=0.35$; AUC threshold-free
 Swept: σ_{res} , σ_{scatt} , T